



Teoler High School

NEET FOUNDATION

REVIEW TEST - 3

Session: 2026 – 27

Grade – IX – D

Time : 3 Hours

Code :

Date : 24-08-2026

Instructions:

- This paper consists of Four parts **Physics, Chemistry, Botany and Zoology**.
- The test is of **3 hr.** duration and the Test Booklet contains **180** multiple-choice questions (four options with a single correct answer) from Physics, Chemistry and Biology (Botany and Zoology).
- Question numbers **1 to 45** are from Physics with one correct answer.
- Question numbers **46 to 90** are from Chemistry with one correct answer.
- Question numbers **91 to 180** are from Biology with one correct answer.
- Four marks will be awarded for each correct answer and one mark will be deducted for each incorrect answer.
- Use Blue/Black Ball Point Pen only for writing particulars on this page/marketing responses on Answer Sheet.
- Rough work is to be done in the space provided for this purpose in the test booklet only.

(Physics)

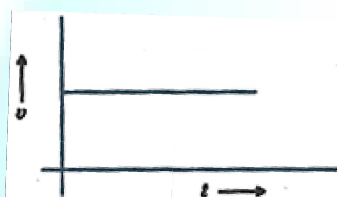
1. A particle is moving in a circular path of radius r . The displacement after half a circle would be:

(1) Zero (2) πr
(3) $2r$ (4) $2\pi r$

2. A body is thrown vertically upward with velocity u , the greatest height h to which it will rise is :

(1) u/g (2) $u^2/2g$
(3) u^2/g (4) $u/2g$

3. From the given $v - t$ graph figure, it can be inferred that the object is :



(1) In uniform motion
(2) At rest
(3) In non-uniform motion
(4) Moving with uniform acceleration

4. Suppose a body is enjoying a ride on a merry-go-round which is moving with a constant speed of 10 ms^{-1} . It implies that the boy is:

(1) At rest
(2) Moving with no acceleration
(3) In acceleration
(4) Moving with uniform velocity

5. Area under a $v - t$ graph represents a physical quantity which has the unit :

(1) m^2 (2) m
(3) m^3 (4) ms^{-1}

6. A ball is thrown vertically upward with an initial velocity of 150 m/s . The ratio of velocity after 3 s and 5 s is

$\frac{x+1}{x}$. The value of the x is [Take, $g = 10 \text{ m/s}^2$]

(1) 10 (2) 5
(3) 6 (4) -5

7. According to the third law of motion, action and reaction :

(1) Always act on the same body
(2) Always act on different bodies in opposite direction
(3) Have same magnitude and directions
(4) Act on either body at normal to each other

8. The inertia of an object tends to cause the object :

(1) To increase its speed
(2) To decrease its speed
(3) To resist any change in its state of motion
(4) To decelerate due to friction

9. A passenger in a moving train tosses a coin which falls behind him. It means that motion of the train is:

(1) Accelerated (2) Uniform
(3) Retarded (4) Along circular tracks

10. The distance covered by a body of mass 5 g having linear momentum 0.3 kg m/s in 5 s is:

(1) 300 m (2) 30 m
(3) 3 m (4) 0.03 m

11. A cricketer catches a ball of mass 150 g in 0.01 s moving with speed 20 m/s . Then he experiences a force of

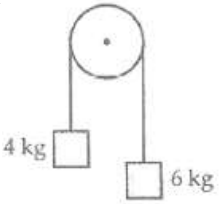
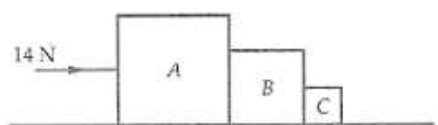
(1) 300 N (2) 30 N
(3) 3 N (4) 0.3 N

12. A force of 6 N acts on a body at rest and of mass 1 kg . During this time, the body attains a velocity of 30 m/s . The time for which the force act on the body is

(1) 7 seconds (2) 5 seconds
(3) 10 seconds (4) 8 seconds

13. A force vector applied on a mass is represented as $\vec{F} = 6\hat{i} - 8\hat{j} + 10\hat{k}$ and accelerates with 1 m/s^2 . What will be the mass of the body?

(1) 10 kg (2) 20 kg
(3) $10\sqrt{2} \text{ kg}$ (4) $2\sqrt{10} \text{ kg}$

14. Sand is being dropped from a conveyor belt at the rate of M kg/s. The force necessary to keep the belt moving with a constant velocity of v m/s will be
 (1) Mv newton (2) $2 Mv$ newton
 (3) $\frac{Mv}{2}$ newton (4) zero
15. A person is standing in an elevator. In which situation, he experiences weight loss?
 (1) When the elevator moves upward with constant acceleration
 (2) When the elevator moves downward with constant acceleration
 (3) When the elevator moves upward with uniform velocity
 (4) When the elevator moves downward with uniform velocity
16. A bullet is fired into a fixed target loses one third of its velocity after travelling 4 cm. It penetrates further $D \times 10^{-3}$ m before coming to rest. The value of D is :
 (1) 32 (2) 3
 (3) 4 (4) 5
17. A stone falls freely such that the distance covered by it in the last second of its motion is equal to the distance covered by it in the first 5 seconds. It remained in air for :
 (1) 12 s (2) 13 s
 (3) 25 s (4) 26 s
18. A bomb moving with velocity $(40\hat{i} + 50\hat{j} - 25\hat{k})\text{ms}^{-1}$ explodes into two pieces of mass ratio 1 : 4. After explosion the small piece move away with velocity $(200\hat{i} + 70\hat{j} + 15\hat{k})\text{ms}^{-1}$. The velocity of larger piece after explosion is
 (1) $45\hat{j} - 35\hat{k}$ (2) $45\hat{i} + 35\hat{j}$
 (3) $45\hat{k} - 35\hat{j}$ (4) $-35\hat{i} + 45\hat{k}$
19. Two bodies of mass 4kg and 6kg are tied to the ends of a massless string. The string passes over a pulley which is frictionless (see figure)
 The acceleration of the system in terms of acceleration due to gravity (g) is
 (1) $g/2$ (2) $g/5$
 (3) $g/10$ (4) g
- 
20. When a ball is thrown vertically up with velocity v_0 , it reaches a maximum height 'h'. If one wishes to triple the maximum height then the ball should be thrown with velocity.
 (1) $\sqrt{3}v_0$ (2) $3v_0$
 (3) $9v_0$ (4) $3/2v_0$
21. Three blocks A, B and C of masses 4kg, 2kg and 1kg respectively, are in contact on a frictionless surface as shown. If a force of 14 N is applied on the 4kg block, then the contact force between A and B is
 (1) 6 N (2) 8 N
 (3) 18 N (4) 2 N
- 
22. A bullet is fired from a gun at the speed of 280 ms^{-1} in the direction 30° above the horizontal. The maximum height attained by the bullet is (Take $g = 9.8\text{ ms}^{-2}$, $\sin 30^\circ = 0.5$)
 (1) 2000 m (2) 1000 m
 (3) 3000 m (4) 2800 m
23. Which of the following is constant in a projectile motion ?
 (1) Horizontal component of the velocity
 (2) Vertical component of the velocity
 (3) Velocity of projection
 (4) All of these
24. At the uppermost point of a projectile, its velocity and acceleration are at an angle of
 (1) 0° (2) 45°
 (3) 90° (4) 180°

25. For a given angle of projection if the initial velocity is doubled the range of the projectile becomes—

(1) half (2) one-fourth
(3) two times (4) four times

26. Two projectiles are thrown with same initial velocity making an angle 45° and 30° with the horizontal respectively. The ratio of their respective ranges will be

(1) $1:\sqrt{2}$ (2) $\sqrt{2}:1$
(3) $2:\sqrt{3}$ (4) $\sqrt{3}:2$

27. For angles of projection of a projectile at angle $(45^\circ - \theta)$ and $(45^\circ + \theta)$, the horizontal range described by the projectile are in the ratio of

(1) $2:1$ (2) $1:1$
(3) $2:3$ (4) $1:2$

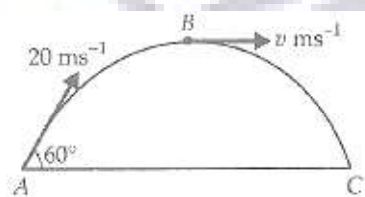
28. A missile is fired for maximum range with an initial velocity of 20 m/s . If $g = 10 \text{ m/s}^2$, then range of the missile is

(1) 40 m (2) 50 m
(3) 60 m (4) 20 m

29. A body is projected horizontally with a velocity of $4\sqrt{2} \text{ m/sec}$. The velocity of the body after 0.7 seconds (Take $g = 10 \text{ m/s}^2$) will be nearly

(1) 10 m/sec (2) 9 m/sec
(3) 19 m/sec (4) 11 m/sec

30. A ball is projected from point A with velocity 20 ms^{-1} at an angle 60° to the horizontal direction. At the highest point B of the path (as shown in figure), the velocity of the ball will be:



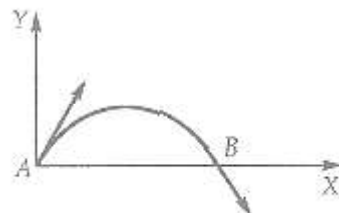
(1) 20 m/s (2) zero
(3) $10\sqrt{3} \text{ m/s}$ (4) 10 m/s

31. A projectile is projected at 30° from horizontal with initial velocity 40 ms^{-1} . The velocity of the projectile at $t = 2 \text{ s}$ from the start will be

(Given : $g = 10 \text{ ms}^{-2}$)

(1) $20\sqrt{3} \text{ ms}^{-1}$ (2) $40\sqrt{3} \text{ ms}^{-1}$
(3) 20 ms^{-1} (4) zero

32. The velocity of a projectile at the initial point A is $(2\hat{i} + 3\hat{j}) \text{ m/s}$. Its velocity (in m/s) at point B is :



(1) $-2\hat{i} - 3\hat{j}$ (2) $-2\hat{i} + 3\hat{j}$
(3) $2\hat{i} - 3\hat{j}$ (4) $2\hat{i} + 3\hat{j}$

33. A stone is projected at angle 30° to the horizontal. The ratio of kinetic energy of the stone at point of projection to its kinetic energy at the highest point of flight will be

(1) $1:2$ (2) $1:4$
(3) $4:1$ (4) $4:3$

34. A particle A is dropped from a height and another particle B is projected in horizontal direction with speed of 5 m/sec from the same height, then correct statement is

(1) particle A will reach the ground first with respect to particle B
(2) particle B will reach the ground first with respect to particle A
(3) both particles will reach the ground simultaneously
(4) both particles will reach the ground with same speed

35. The range of the projectile projected at an angle of 15° with horizontal is 50 m . If the projectile is projected with same velocity at an angle of 45° with horizontal, then its range will be

(1) 50 m (2) $50\sqrt{2} \text{ m}$
(3) 100 m (4) $100\sqrt{2} \text{ m}$

36. The angle between the two vectors

$$\vec{A} = 3\hat{i} + 4\hat{j} + 5\hat{k} \text{ and } \vec{B} = 3\hat{i} + 4\hat{j} - 5\hat{k} \text{ will be}$$

- (1) 90° (2) 180°
(3) zero (4) 45°

37. If $|\vec{A} \times \vec{B}| = \sqrt{3} \vec{A} \cdot \vec{B}$, then the value of $|\vec{A} + \vec{B}|$ is

- (1) $(A^2 + B^2 + AB)^{1/2}$
(2) $\left(A^2 + B^2 + \frac{AB}{\sqrt{3}}\right)^{1/2}$
(3) $A + B$
(4) $(A^2 + B^2 + \sqrt{3}AB)^{1/2}$

38. If $\vec{F} = 2\hat{i} + \hat{j} - \hat{k}$ and $\vec{r} = 3\hat{i} + 2\hat{j} - 2\hat{k}$, then the scalar and vector products of $\vec{F}$ and $\vec{r}$ have the magnitudes respectively as

- (1) $5, \sqrt{3}$ (2) $4, \sqrt{5}$
(3) $10, \sqrt{2}$ (4) $10, 2$

39. $\vec{A}$ is a vector quantity such that $|\vec{A}| = \text{non-zero constant}$. Which of the following expression is true for $\vec{A}$?

- (A) $\vec{A} \cdot \vec{A} = 0$ (2) $\vec{A} \times \vec{A} < 0$
(3) $\vec{A} \times \vec{A} = 0$ (4) $\vec{A} \times \vec{A} > 0$

40. If $\vec{A}$ and $\vec{B}$ are two vectors satisfying the relation $\vec{A} \cdot \vec{B} = |\vec{A} \times \vec{B}|$, then the value of $|\vec{A} - \vec{B}|$ will be

- (1) $\sqrt{A^2 + B^2}$ (2) $\sqrt{A^2 + B^2 - \sqrt{2} AB}$
(3) $\sqrt{A^2 + B^2 + 2 AB}$ (4) $\sqrt{A^2 + B^2 + \sqrt{2} AB}$

41. If two vectors $\vec{P} = \hat{i} + 2m\hat{j} + m\hat{k}$ and $\vec{Q} = 4\hat{i} - 2\hat{j} + m\hat{k}$ are perpendicular to each other, then the value of m will be

- (1) 1 (2) -1
(3) -3 (4) 2

42. Two vectors $\vec{A}$ and $\vec{B}$ are perpendicular if :

- (1) $\vec{A} \cdot \vec{B} = 1$ (2) $\vec{A} \times \vec{B} = 0$
(3) $\vec{A} \cdot \vec{B} = 0$ (4) $\vec{A} \cdot \vec{B} = AB$

43. The vector $\vec{A}$ and $\vec{B}$ are such that $|\vec{A} + \vec{B}| = |\vec{A} - \vec{B}|$. The angle between the two vectors is

- (1) 45° (2) 90°
(3) 60° (4) 75°

44. A vehicle travels half the distance with speed v and the remaining distance with speed $2v$. Its average speed is :

- (1) $\frac{2v}{3}$ (2) $\frac{4v}{3}$
(3) $\frac{3v}{4}$ (4) $\frac{v}{3}$

45. A bus travelling the first one-third distance at a speed of 10 km/h, the next one-third at 20 km/h and the last one-third at 60 km/h. The average speed of bus is:

- (1) 9 km/h (2) 16 km/h
(3) 18 km/h (4) 48 km/h

(Chemistry)

46. Number of oxygen atoms in 8 g of ozone is -

- (1) 6.02×10^{23} (2) $\frac{6.02 \times 10^{23}}{2}$
(3) $\frac{6.02 \times 10^{23}}{3}$ (4) $\frac{6.02 \times 10^{23}}{6}$

47. If 0.5 mol of BaCl_2 is mixed with 0.2 mol of Na_3PO_4 , the maximum number of moles of $\text{Ba}_3(\text{PO}_4)_2$ that can be formed is -

- $3\text{BaCl}_2 + 2\text{Na}_3\text{PO}_4 \longrightarrow \text{Ba}_3(\text{PO}_4)_2 + 6\text{NaCl}$
(1) 0.7 (2) 0.5
(3) 0.3 (4) 0.1

48. 10 g of hydrogen and 64 g of oxygen were filled in a steel vessel and exploded. Amount of water produced in this reaction will be :-
 (1) 1 mol (2) 2 mol
 (3) 3 mol (4) 4 mol
49. A mixture of gases contains H_2 and O_2 gases in the ratio of 1 : 4 (w/w). What is the molar ratio of the two gases in the mixture ?
 (1) 4 : 1 (2) 16 : 1
 (3) 2 : 1 (4) 1 : 4
50. Equal masses of H_2 , O_2 and methane have been taken in a container of volume V at temperature $27^\circ C$ at identical conditions. The ratio of the volumes of gases H_2 : O_2 : CH_4 would be :
 (1) 8 : 16 : 1 (2) 16 : 8 : 1
 (3) 16 : 1 : 2 (4) 8 : 1 : 2
51. When 22.4 L of $H_2(g)$ is mixed with 11.2 L of $Cl_2(g)$ at S.T.P., the moles of $HCl(g)$ formed is equal to:-
 (1) 1 mol of $HCl(g)$ (2) 2 mol of $HCl(g)$
 (3) 0.5 mol of $HCl(g)$ (4) 1.5 mol of $HCl(g)$
52. Which pair of the gaseous species diffuse through a small jet with the same rate of diffusion at same P and T:
 (1) NO, CO (2) NO, CO_2
 (3) NH_3 , PH_3 (4) NO, C_2H_6
53. The increasing order of effusion among the gases, H_2 , O_2 , NH_3 and CO_2 is –
 (1) H_2 , CO_2 , NH_3 , O_2 (2) H_2 , NH_3 , O_2 , CO_2
 (3) H_2 , O_2 , NH_3 , CO_2 (4) CO_2 , O_2 , NH_3 , H_2
54. A balloon filled with methane CH_4 is pricked with a sharp point and quickly plunged into a tank of hydrogen at the same pressure. After sometime the balloon will have :
 (1) Enlarged
 (2) Collapsed
 (3) Remain unchanged in size
 (4) Ethylene (C_2H_4) inside it
55. A bottle of dry ammonia and a bottle of dry hydrogen chloride connected through a long tube are opened simultaneously at both ends, the white ammonium chloride ring first formed will be :
 (1) at the centre of the tube
 (2) near the hydrogen chloride bottle
 (3) near the ammonia bottle
 (4) throughout the length of the tube
56. The minimum pressure required to compress 600 dm^3 of a gas at 1 bar to 150 dm^3 at $40^\circ C$ is
 (1) 4.0 bar (2) 0.2 bar
 (3) 1.0 bar (4) 2.5 bar
57. Choose the correct option for the total pressure (in atm.) in a mixture of 4g O_2 and 2g H_2 confined in a total volume of one litre at $0^\circ C$ is:
 [Given $R = 0.082\text{ L atm mol}^{-1}\text{K}^{-1}$, $T = 273\text{K}$]
 (1) 2.518 (2) 2.602
 (3) 25.18 (4) 28.02
58. A mixture of N_2 and Ar gases in a cylinder contains 7g of N_2 and 8g of Ar. If the total pressure of the mixture of gases in the cylinder is 27 bar, the partial pressure of N_2 is:
 [Use atomic masses (in g mol^{-1}) : N = 14, Ar = 40]
 (1) 18 bar (2) 9 bar
 (3) 12 bar (4) 15 bar
59. A certain gas takes three times as long to effuse out as helium. Its molecular mass will be :
 (1) 64 u (2) 9 u
 (3) 27 u (4) 36 u
60. In water saturated air, the mole fraction of water vapour is 0.02. If the total pressure of the saturated air is 1.2 atm, the partial pressure of dry air is :-
 (1) 3.18 atm (2) 4.76 atm
 (3) 1.176 atm (4) 2.98 atm
61. The volume occupied by 1.8 g of water vapour at $374^\circ C$ and 1 bar pressure will be :-
 [Use $R = 0.083\text{ bar L K}^{-1}\text{ mol}^{-1}$]
 (1) 96.66 L (2) 55.87 L
 (3) 3.10 L (4) 5.37 L

62. Two gases A and B having the same volume diffuse through a porous partition in 20 and 10 seconds respectively. The molecular mass of A is 49u. Molecular mass of B will be :-
 (1) 50.00 u (2) 12.25 u
 (3) 6.50 u (4) 25.00 u
63. The densities of two gases are in the ratio of 1 : 16. The ratio of their rates of diffusion is :
 (1) 16 : 1 (2) 4 : 1
 (3) 1 : 4 (4) 1 : 16
64. The rate of diffusion of a gas having molecular weight just double of nitrogen gas is 56 ml per sec the rate of diffusion of nitrogen gas will be :
 (1) 79.19 ml/sec (2) 112 ml/sec
 (3) 56 ml/sec (4) 90 ml/sec
65. If rate of diffusion of A is 5 times that of B. What will be the ratio of density of A and B :
 (1) 1/25 (2) 1/5
 (3) 25 (4) 5
66. Since the atomic weights of carbon, nitrogen and oxygen are 12, 14 and 16 respectively, among the following pairs of gases, the pair that will diffuse at the same rate is :
 (1) Carbon dioxide (CO₂) and nitrous oxide (N₂O)
 (2) Carbon dioxide (CO₂) and nitrogen dioxide (NO₂)
 (3) Carbon dioxide (CO₂) and carbon monoxide (CO)
 (4) Carbon dioxide (CO₂) and nitric oxide (NO)
67. Moles of oxygen atoms in 40 gm of SO₃ will be:
 (1) 0.3 mole (2) 15 mole
 (3) 1.5 mole (4) 3 mole
68. 1 gm-atom of nitrogen may represents :
 (1) 6.02×10^{23} N₂ molecules
 (2) 22.4 lit. of N₂ at 1 atm & 273 K
 (3) 11.2 lit. of N₂ at 1 atm & 0°C
 (4) 28 g of nitrogen
69. The number of neutrons present in 9 mg of O¹⁸ is
 (1) 10 (2) 5N_A
 (3) 0.005 N_A (4) 0.0005 N_A
70. Which of the following have largest number of atoms?
 (1) 4 gm oxygen (2) 16 gm sulphur
 (3) 35.5 gm chlorine (4) 14 gm lithium
71. 4.6×10^{22} atoms of an element weigh 13.8 gm. The gram atomic mass of the element is:
 (Take N_A = 6×10^{23})
 (1) 120 gm (2) 180 gm
 (3) 35.5 gm (4) 108 gm
72. Which of the following contain largest number of total atoms?
 (1) 1 gm atom carbon (2) 1 gm oxygen atoms
 (3) 3.4 gm H₂S (4) 1.6 gm CH₄
73. If mass of 2 atoms is 4×10^{-23} gm. Then atomic mass of element will be :
 [Take : N_A = 6×10^{23} / mole]
 (1) 24 (2) 12
 (3) 6 (4) 2
74. Which of the following is a pure-substance?
 (1) Air (2) Distilled water
 (3) Steel (4) Brass
75. Pure water contains hydrogen atoms and oxygen atom which cannot be separated by physical methods are in which ratio?
 (1) 3 : 1 (2) 2 : 1
 (3) 1 : 1 (4) 1 : 4
76. Which of the following is not a pure substance?
 (1) Baking soda (2) Diamond
 (3) Sea water (4) Table salt
77. The process used to separate oil and water is
 (1) distillation (2) sublimation
 (3) chromatography (4) separating funnel
78. Which of the following is/are pure substance?
 (1) oxygen (2) sulphur
 (3) iron (4) all of these
79. Filtration can be used to separate:
 (1) solids from solids (2) liquids from gases
 (3) liquids from liquids (4) liquids from solids

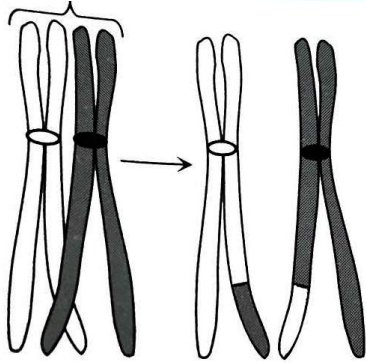
80. Bronze is an alloy of
 (1) Copper and Zinc (2) Copper and Tin
 (3) Zinc and Iron (4) Copper and Gold
81. Stainless steel is an alloy of
 (1) Copper
 (2) Iron, nickel and chromium
 (3) Manganese
 (4) Copper and Gold
82. From the below options, choose the correct example for gaseous solutions.
 (1) Oxygen dissolved in water
 (2) Camphor in nitrogen gas
 (3) Carbon dioxide dissolved in water
 (4) Hydrogen in palladium
83. A jug contains 2L of milk. Calculate the volume of the milk in m^3 .
 (1) $4 \times 10^{-3} \text{ m}^3$ (2) $2 \times 10^{-3} \text{ m}^3$
 (3) $4 \times 10^3 \text{ m}^3$ (4) $2 \times 10^3 \text{ m}^3$
84. How many moles of methane are required to produce 22g $\text{CO}_2(\text{g})$ after combustion?
 (1) 0.5 (2) 2
 (3) 1.5 (4) 3
85. Find number of protons in 180 ml H_2O . Density of water = 1 gm/ml.
 (1) 3.011×10^{-25} protons
 (2) 3.011×10^{25} protons
 (3) 6.022×10^{-25} protons
 (4) 6.022×10^{25} protons
86. On a ship sailing in Pacific Ocean where temperature is 23.4°C , a balloon is filled with 2L air. What will be the volume of the balloon when the ship reaches Indian ocean, where temperature is 26.1°C ?
 (1) 2.018 L (2) 0.218 L
 (3) 20.18 L (4) 201.8 L
87. Aluminium reacts with sulphur to form aluminium sulphide. If 5.4 gm of Aluminium reacts with 12.8 gm sulphur gives 12 gm of aluminium sulphides, then the percent yield of the reaction is. (Al = 27, S = 32)
 (1) 66.67% (2) 40%
 (3) 20% (4) 80%
88. One gram molecules of a gas at N.T.P. occupies 22.4 litres. The fact was derived from
 (1) Dalton's theory
 (2) Avogadro's hypothesis
 (3) Berzelius hypothesis
 (4) Law of gaseous volume
89. A vessel has 6g of oxygen at a pressure P and temperature 400 K. A small hole is made in it so that O_2 leaks out. How much O_2 leaks out if the pressure is P/2 and temperature 300 K.
 (1) 5g (2) 4g
 (3) 2g (4) 3g
90. What mass of sodium chloride would be decomposed by 9.8 g of sulphuric acid, if 12g of sodium bisulphate and 2.75 g of hydrogen chloride were produced in a reaction assuming that the law of conservation of mass is true?
 (1) 7.35 g (2) 12.55 g
 (3) 4.95 g (4) 8.75 g
- (Botany)**
91. Which statement correctly describes the principal function of the Golgi apparatus?
 (1) It packages materials for delivery within the cell or secretion outside the cell.
 (2) It synthesises proteins directly from amino acids.
 (3) It produces ATP during aerobic respiration.
 (4) It stores the genetic material of the cell.
92. Which property of the plasma membrane enables it to regulate the movement of substances across the cell boundary?
 (1) It is composed entirely of proteins.
 (2) It is completely impermeable to polar molecules.
 (3) It is selectively permeable.
 (4) It contains a rigid layer of cellulose.

93. Which combination correctly represents the constituents of chromatin?
 (1) DNA, histones, non-histone proteins and RNA
 (2) DNA, cellulose, proteins and lipids
 (3) RNA, phospholipids and carbohydrates only
 (4) DNA and histones only
94. Which sequence correctly represents the layers of a typical bacterial cell envelope from the outermost to the innermost side?
 (1) Cell wall → glycocalyx → plasma membrane
 (2) Plasma membrane → cell wall → glycocalyx
 (3) Glycocalyx → cell wall → plasma membrane
 (4) Glycocalyx → plasma membrane → cell wall
95. A substance is transported across the plasma membrane from a region of lower concentration to a region of higher concentration. Which statement best describes this process?
 (1) It is osmosis and involves only the movement of water.
 (2) It is active transport and requires energy supplied through ATP.
 (3) It is simple diffusion and does not require energy.
 (4) It is passive transport occurring along the concentration gradient.
96. On what basis are bacteria classified as Gram-positive or Gram-negative?
 (1) The number and arrangement of their flagella
 (2) Differences in their cell envelopes and their response to Gram staining
 (3) The presence or absence of plasmid DNA
 (4) The shape and size of their ribosomes
97. What is the main role of plasmodesmata in plant tissues?
 (1) They connect the cytoplasm of neighbouring cells.
 (2) They attach chromosomes to spindle fibres.
 (3) They connect the thylakoids of different grana.
 (4) They join the inner and outer mitochondrial membranes.
98. During which stages of mitosis and meiosis, respectively does the centromere of each chromosome split?
 (1) Metaphase, Metaphase II
 (2) Prophase, Telophase II
 (3) Telophase, Anaphase I
 (4) Anaphase, Anaphase II
99. Among eukaryotes, replication of DNA takes place in:
 (1) S Phase (2) G1 Phase
 (3) G2 Phase (4) M Phase
100. Which one of the following never occurs during mitotic cell division?
 (1) Spindle fibres attach to kinetochores of chromosomes.
 (2) Movement of centrioles towards opposite poles.
 (3) Pairing of homologous chromosomes.
 (4) Coiling and condensation of chromosomes.
101. Consider the following statements about cell division and choose the correct option:
Statement I : Mitosis produces identical cells for body growth.
Statement II : Meiosis halves chromosome numbers to form gametes.
 (1) Both Statement I and Statement II are correct.
 (2) Both Statement I and Statement II are incorrect.
 (3) Statement I is correct but Statement II is incorrect.
 (4) Statement I is incorrect but Statement II is correct.
102. The fruit fly has 8 chromosomes ($2n$) in each cell. During interphase of Mitosis if the number of chromosomes at G1 phase is 8, what would be the number of chromosomes after S phase?
 (1) 32 (2) 8
 (3) 16 (4) 4
103. Cells in G_0 phase:
 (1) suspend the cell cycle
 (2) terminate the cell cycle
 (3) exit the cell cycle
 (4) enter the cell cycle

104. The enzyme recombinase is required at which stage of meiosis?

- (1) Pachytene (2) Zygotene
(3) Diplotene (4) Diakinesis

105. Given below is the representation of a certain event at a particular stage of a type of cell division. Which is this stage?



- (1) Prophase of mitosis
(2) Both prophase and metaphase of mitosis
(3) Prophase I during Meiosis I
(4) Prophase II during Meiosis

106. If you are provided with root-tips of onion in your class and are asked to count the chromosomes, which of the following stages can you most conveniently look into?

- (1) Metaphase (2) Telophase
(3) Anaphase (4) Prophase

107. Mitotic spindle is mainly composed of which protein?

- (1) Actin (2) Myosin
(3) Tubulin (4) Keratin

108. **Assertion (A):** Interphase is resting stage.

Reason (R): The interphase cell is metabolically inactive.

- (1) If both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) If both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(3) If Assertion is true but Reason is false.
(4) If both Assertion and Reason are false.

109. Which of the following is a characteristic of cytokine-

sis in plant cells?

- (1) Cell furrow (2) Cell plate
(3) Cell wall (4) None of these

110. **Assertion (A):** Meiosis I is reductional division.

Reason (R): Chromosome number reduces to half.

- (1) If both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) If both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(3) If Assertion is true but Reason is false.
(4) If both Assertion and Reason are false.

111. Collenchyma shows deposition of:

- (1) Pectin (2) Suberin
(3) Resin (4) Lignin

112. Which cells regulate the function of sieve tube?

- (1) Guard cells (2) Passage cells
(3) Companion cells (4) Bulliform cells

113. Which of the following cells is living but is without nucleus?

- (1) Sieve cells (2) Companion cells
(3) Sieve tube (4) Guard cells

114. Which of the following is absent in the phloem of monocots?

- (1) Companion cells (2) Phloem parenchyma
(3) Sieve cells (4) Phloem fibre

115. Match Columns I with II.

I	II
1. Parenchyma	(a) Increase in length of plants
2. Lateral meristem	(b) Increase in nodal region
3. Apical meristem	(c) Support, protection and storage
4. Intercalary meristem	(d) Increase in girth

- (1) 1-d, 2-c, 3-b, 4-a (2) 1-c, 2-d, 3-a, 4-b
(3) 1-a, 2-b, 3-d, 4-c (4) 1-c, 2-a, 3-d, 4-b

116. Which of the following is not a simple tissue?

- (1) Xylem (2) Parenchyma
(3) Collenchyma (4) Sclerenchyma

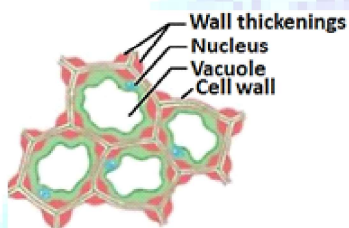
117. Water conducting elements of gymnosperms and primitive angiosperms are:

- (1) Vessels (2) Vessel elements
(3) Tracheids (4) Fibrotracheids

118. Sugar transport elements of gymnosperms and pteridophytes are:

- (1) Sieve cells (2) Sieve elements
(3) Sieve tubes (4) Sieve tube elements

119. The main function of the plant tissue shown below is:



- (1) Storage of nutrients
(2) Providing buoyancy to submerged plants
(3) Mechanical support to growing parts of a plant
(4) Increase in length of the plant

120. Which among the following is wrong about Sclerenchyma?

- (1) These are the living tissues responsible for stiffness in plants
(2) The husk of a coconut is made of Sclerenchymatous tissue
(3) They are of two kinds, namely, sclereids and fibers
(4) Simple pits are present in thickened cell walls of sclerenchyma

121. Vascular cambium and cork cambium are examples of _____.

- (1) Apical meristem
(2) Intercalary meristem
(3) Lateral meristem
(4) Permanent tissue

122. Permanent tissues originate from _____.

- (1) Apical meristem
(2) Lateral meristem
(3) Differentiation of meristematic cells
(4) Root hairs

123. Parenchyma with large air spaces is called _____.

- (1) Collenchyma (2) Aerenchyma
(3) Chlorenchyma (4) Sclerenchyma

124. Gritty texture in guava/pear is due to _____.

- (1) Fibres (2) Sclereids
(3) Parenchyma (4) Collenchyma

125. Which cells in the vascular bundle provide structural support?

- (1) Xylem vessels (2) Xylem fibers
(3) Companion cells (4) Phloem parenchyma

126. Cells of permanent tissue are specialised:

- (1) Only functionally.
(2) Only structurally.
(3) Both functionally and structurally.
(4) For meristematic activity.

127. Tracheids differ from vessels in:

- (1) Having lignified cell wall.
(2) Absence of protoplast.
(3) Having perforations in their common walls.
(4) Possessing oblique and tapering ends.

128. Which component of the xylem listed here is involved in radial conduction of water?

- (1) Tracheids (2) Xylem vessels
(3) Xylem parenchyma (4) Xylem fibres

129. Which part of the plant provides us with industrial level jute fibres?

- (1) Interxylary fibres (2) Xylem fibres
(3) Phloem fibres (4) None of these

130. How does ground grass and Bamboo elongate itself?

- (1) Apical meristem (2) Lateral meristem
(3) Secondary meristem (4) Intercalary meristem

131. Which tissue is called Universal tissue?

- (1) Xylem (2) Parenchyma
(3) Collenchyma (4) Sclerenchyma

132. Identify the structure labelled in the given tissue:

- (1) Perforation plate
- (2) Sieve plate
- (3) Disc plate
- (4) All of the above



133. Exarch xylem condition is found in _____.

- (1) Stem (2) Root
- (3) Leaf (4) Pith

134. Which of the following xylem elements does not play a direct role in conduction of water?

- (1) Tracheids (2) Vessels
- (3) Xylem parenchyma (4) Xylem fibers

135. Which type of parenchyma is modified for photosynthesis?

- (1) Storage parenchyma (2) Chlorenchyma
- (3) Aerenchyma (4) Prosenchyma

ZOOLOGY

136. Which is an auto-immune disease?

- (1) Cancer
- (2) Polio
- (3) Erythroblastosis foetalis
- (4) AIDS

137. Microbes that generally enter the body through the nose are likely to affect

- (1) Gut (2) Lungs
- (3) Liver (4) Lymph nodes

138. The type of disease that begins abruptly, affects for a short duration is termed as:

- (1) Chronic Disease (2) Acute Disease
- (3) Hormonal Disease (4) Deficiency disease

139. The droplet method of transmission of the disease is found in:

- (1) Common cold (2) AIDS
- (3) Hepatitis (4) Polio

140. Which of the following diseases is NOT transmitted through a mosquito bite?

- (1) Malaria (2) Dengue
- (3) Amoebiasis (4) Filariasis

141. Which type of cells produce antibodies in the human body?

- (1) T-lymphocytes (2) B-lymphocytes
- (3) Macrophages (4) Neutrophils

142. Which of the following statements about cell-mediated immunity is correct?

- (1) Cell-mediated immunity is provided by B-lymphocytes.
- (2) Cytotoxic T-cells play a key role in cell-mediated immunity.
- (3) Cell-mediated immunity involves the production of antibodies.
- (4) Cell-mediated immunity does not target virus-infected cells.

143. Which of the following is NOT a bacterial disease?

- (1) Typhoid (2) Pneumonia
- (3) Common cold (4) Diphtheria

144. What is the primary function of lymph nodes in the immune system?

- (1) Production of red blood cells
- (2) Filtration of lymph and activation of immune cells
- (3) Secretion of digestive enzymes
- (4) Production of platelets

145. Which of the following statements about interferons is TRUE?

- (1) They are produced by bacteria
- (2) They are used for treating bacterial infections
- (3) They help protect cells from viral infections
- (4) They are a type of antibody

146. Which cells act as the first line of defense in innate immunity?

- (1) B-lymphocytes (2) T-lymphocytes
- (3) Macrophages (4) Mast cells

147. Which of the following statements about antibodies is correct?
- (1) Antibodies are produced by T-lymphocytes.
 - (2) Plasma cells secrete antibodies.
 - (3) Antibodies directly kill pathogens.
 - (4) Antibodies are part of the innate immune system.
148. Which of the following statements about immunoglobulins (antibodies) is correct?
- (1) IgG is the most abundant antibody in blood plasma.
 - (2) IgM is responsible for allergic reactions.
 - (3) IgE provides long-term immunity.
 - (4) IgA is primarily found in the bloodstream.
149. Which type of vaccine contains inactivated or killed pathogens?
- (1) Live attenuated vaccine
 - (2) Toxoid vaccine
 - (3) Killed vaccine
 - (4) Recombinant vaccine
150. Which of the following is the most common mode of transmission of tuberculosis?
- (1) Contaminated water
 - (2) Direct skin contact
 - (3) Inhalation of airborne droplets
 - (4) Vector transmission
151. Which of the following is an example of passive immunity?
- (1) Vaccination
 - (2) Injection of pre-formed antibodies
 - (3) Natural infection and recovery
 - (4) Development of memory cells
152. Which of the following substances is released by mast cells during an allergic reaction?
- (1) Histamine
 - (2) Insulin
 - (3) Dopamine
 - (4) Melatonin
153. Which mosquito species is responsible for transmitting dengue fever?
- (1) *Culex*
 - (2) *Anopheles*
 - (3) *Aedes*
 - (4) *Mansonia*
154. In which blood corpuscles, the HIV undergoes replication and produces progeny viruses?
- (1) B- lymphocytes
 - (2) Basophils
 - (3) Eosinophils
 - (4) T_H Cells
155. Identify the correct pair representing the causative agent of typhoid fever and the confirmatory test for typhoid.
- (1) *Salmonella typhi*- Anthorone test
 - (2) *Salmonella typhi*- Widal test
 - (3) *Plasmodium vivax*- UTI test
 - (4) *Streptococcus pneumoniae*- Widal test
156. Pseudostratified epithelium is present in:
- (1) nephron and neuron
 - (2) larynx and pharynx
 - (3) trachea and bronchi
 - (4) urinary bladder and intestine
157. Columnar epithelium with microvilli or Brush Border is present in:
- (1) Gall bladder
 - (2) Stomach
 - (3) Appendix
 - (4) Pharynx
158. The internal lining of blood vessels is called as:
- (1) Mesothelium
 - (2) Endothelium
 - (3) Pavement Epithelium
 - (4) Stratified Epithelium
159. Germinal epithelium of ovary is formed of:
- (1) Columnar epithelium
 - (2) Squamous epithelium
 - (3) Cuboidal epithelium
 - (4) Stratified epithelium
160. Epidermis of skin of vertebrates comprises of:
- (1) Simple epithelium
 - (2) Stratified epithelium
 - (3) Transitional epithelium
 - (4) Columnar epithelium
161. Mucus cells (Goblet cells) are:
- (1) Unicellular gland
 - (2) Multicellular glands
 - (3) Endocrine glands
 - (4) Parietal cells of gastric glands

162. Desmosomes:

- (1) connect epithelial cells
- (2) are types of lysosomes
- (3) are granules of muscle fibres
- (4) are found in bone

163. Blood colloidal osmotic pressure is maintained by:

- (1) Albumin (2) Globulin
- (3) Fibrinogen (4) Thrombin

164. **Assertion (A):** Blood has property of clotting.

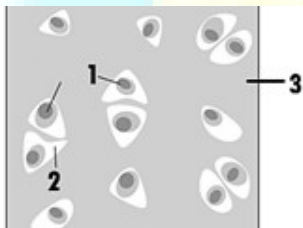
Reason (R): Blood has plasma protein 'Fibrinogen'.

- (1) If both Assertion and Reason are true and Reason is the correct explanation of Assertion.
- (2) If both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
- (3) If Assertion is true but Reason is false.
- (4) If both Assertion and Reason are false.

165. This one is the characteristic of epithelial tissue:

- (1) Tissues are highly vascularized
- (2) These cells never produce glands
- (3) The cells will have a rapid rate of cell division
- (4) Large intercellular spaces are seen between cells

166. Which of the following is correct for labels 1, 2 and 3 in the given diagram?



- (1) 1-Lacunae, 2-Matrix, 3- Chondrocyte
- (2) 1-Chondrocyte, 2- Lacunae, 3-Matrix
- (3) 1- Chondrocyte, 2- Lacunae, 3- Elastic fibre
- (4) 1-Matrix, 2- Chondrocytes, 3- Lacunae

167. Blood plasma is:

- (1) Neutral (2) Slightly acidic
- (3) Slightly alkaline (4) Strongly acidic

168. Which of the following is a transparent tissue?

- (1) Tendon (2) Hyaline cartilage
- (3) Fibrous cartilage (4) All of these

169. Lacuna of bone contains:

- (1) One osteocyte (2) Two osteocytes
- (3) Many osteocytes (4) Many chondrocytes

170. Major protein of connective tissue is:

- (1) Myosin (2) Melanin
- (3) Collagen (4) Keratin

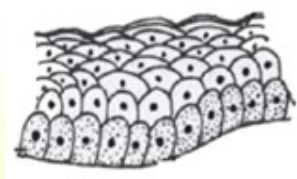
171. Light bands of muscle fibre are made of the protein:

- (1) Tubulin (2) Myosin
- (3) Actin (4) Myoglobin

172. Canaliculi are found associated with:

- (1) Cartilage (2) Bone
- (3) Muscle (4) Neuron

173. Which tissue is indicated in the given figure?



- (1) Pseudostratified (2) Ciliated
- (3) Cuboidal (4) Stratified

174. **Assertion (A):** Skeletal muscle is also known as voluntary muscle.

Reason (R): They can contract according to desire of the animal.

- (1) If both Assertion and Reason are true and Reason is the correct explanation of Assertion.
- (2) If both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
- (3) If Assertion is true but Reason is false.
- (4) If both Assertion and Reason are false.

175. It connects the bones joints and holds them in position:

- (1) Tendon (2) While elastic cartilage
- (3) Ligament (4) (1) and (2) both

176. The outer covering of cartilage is called:
- (1) Peritoneum (2) Periosteum
(3) Endosteum (4) Perichondrium
177. Which type of epithelium is involved in a function to move particles or mucus in specific direction?
- (1) Squamous epithelium (2) Cuboidal epithelium
(3) Columnar epithelium (4) Ciliated epithelium
178. When the surface cells of stratified epithelium contain insoluble protein (Keratin) the tissue is called :-
- (1) Stratified Squamous Keratinised
(2) Stratified Ciliated Keratinised
(3) Stratified Cuboidal Keratinised
(4) Stratified Columnar Keratinised
179. Which of the following tissue is normally found in tendon?
- (1) Hyaline cartilage (2) White fibrous tissue
(3) Ligament (4) Areolar tissue
180. Multi-lobed nucleus and granular cytoplasm are characteristics of which of the WBCs?
- (1) Neutrophil (2) Monocytes
(3) Lymphocytes (4) Eosinophils

* * * * *

Answer Key

Grade : IX-D

Review Test-3

Date : 24-8-2026

CODE : 23526 & 28659

PHYSICS				CHEMISTRY				BOTANY				ZOOLOGY			
Q.No.	1	Ans.	3	Q.No.	46	Ans.	2	Q.No.	91	Ans.	1	Q.No.	136	Ans.	3
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